CLUBSPHERE: SMART CLUB MEMBERSHIP, FINANCE & EVENT ANALYTICS SYSTEM

MINI PROJECT REPORT

24MNC201 ESSENTIALS OF OFFICE AUTOMATION

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BONA FIDE CERTIFICATE

This is to certify that the Mini Project Report entitled “CLUBSPHERE: SMART CLUB MEMBERSHIP, FINANCE & EVENT ANALYTICS SYSTEM” submitted by Asif Hussain (AJC25CS058), in partial fulfillment of the requirement for the award of the Bachelor of Technology in Computer Science and Engineering is a bona fide record of the work carried out at AMAL JYOTHI COLLEGE OF ENGINEERING (AUTONOMOUS), under our guidance and supervision.

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ABSTRACT

College clubs must manage member records, fee collection, events and attendance, but these tasks are often handled in separate lists that are slow to update and easy to get wrong. This mini project presents ClubSphere, a Smart Club Membership, Finance and Event Analytics System built entirely in Microsoft Excel without macros or programming. The workbook links member, fee, event and attendance records through unique Member IDs and Event IDs, and formulas such as EDATE, IF, COUNTIFS, SUMIFS and INDEX with MATCH calculate renewal dates, membership status, fee balances and attendance percentages automatically. Data validation, dropdown lists and conditional formatting reduce entry errors and highlight overdue or expiring records. A Member Profile sheet, a Reports sheet and a Dashboard with key indicators, alerts, charts and leaderboards present the results and update whenever a record changes. Testing with fictional data of 40 members, 52 fee records and 7 events showed that all calculations responded correctly to changes. An extension module built with Microsoft Word mail merge uses the same data to produce 40 membership confirmation letters, 17 renewal reminders, 16 fee notices and 40 ID cards without retyping. The merged documents were checked against the data source and saved as Word and PDF files. Together the two parts show that built-in office tools alone can deliver a practical administration and communication system for a small organisation.

Keywords: Microsoft Excel, Microsoft Word, Mail Merge, Office Automation, Club Management, Dashboard, Data Validation

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LIST OF SYMBOLS AND ABBREVIATIONS

Abbreviation	Meaning
KPI	Key Performance Indicator
UI	User Interface
ID	Identifier
VBA	Visual Basic for Applications
Rs.	Indian Rupee
CSE, ECE, EEE, ME, CE, MCA	Computer Science, Electronics and Communication, Electrical and Electronics, Mechanical, Civil Engineering, Master of Computer Applications

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Chapter 1
Introduction

1.1 Background

A college club is a small organisation with many connected activities. It registers members, collects membership and event fees, renews memberships every year, conducts events and records who attended them. When each of these activities is kept in a separate list or paper register, a coordinator needs a lot of time to answer simple questions such as who has paid, whose membership has expired, or how many members attended an event.

Office automation software offers a practical way to organise such records. Spreadsheets in particular are widely available and allow data to be arranged in tables, calculated with formulas and summarised with charts (Alexander, Kusleika and Walkenbach, 2018). This mini project uses Microsoft Excel to build ClubSphere, a Smart Club Membership, Finance and Event Analytics System, which links the records of a club and shows the results on one dashboard. Figure 1.1 gives an overview of the idea.

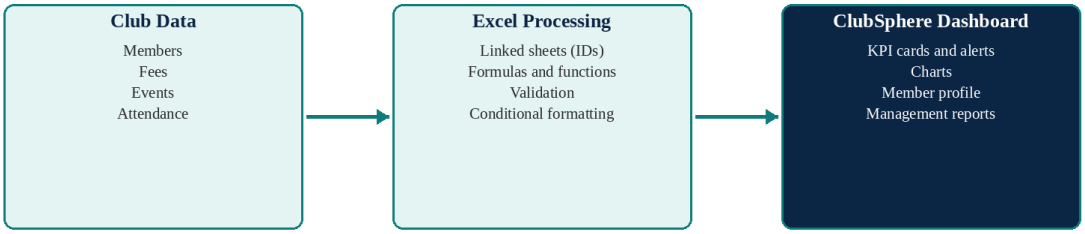


Figure 1.1: Overview of the ClubSphere system

1.2 Motivation

Manual record keeping has three common problems: the same information is typed more than once, updates in one list are not reflected in the others, and reports take a long time to prepare. These problems grow as the number of members and events increases. A connected workbook in which a change in one place automatically updates every dependent result removes much of this effort and reduces mistakes.

The project was also chosen to demonstrate practical office automation skills: structured tables, lookup and summary functions, data validation, conditional formatting, charts and page layout, all without macros or programming.

1.3 Scope of the Project

ClubSphere is a single Excel workbook for one club. It manages members, fees, events and attendance, calculates statuses and totals, and presents analytics. It is built for up to 100

members, 150 fee records, 20 events and 300 attendance records and is demonstrated with fictional sample data. Automatic emails, a web portal, login systems and database integration are outside the scope because they need programming or external services. As an extension, the same membership data is used with Microsoft Word mail merge to produce personalised letters and ID cards (Chapter 6).

1.4 Organisation of the Report

Chapter 2 reviews the background and defines the problem, gap, objectives and constraints. Chapter 3 describes the user interface with screenshots. Chapter 4 explains the implementation, formulas and validation rules. Chapter 5 presents testing and results, Chapter 6 describes the Word mail merge extension, and Chapter 7 gives the conclusion and future scope.

Chapter 2
Literature Review

2.1 Spreadsheets for Record Keeping

When every row of a sheet represents one record and every record has a unique identifier, a spreadsheet can behave like a simple database. Lookup functions retrieve related data using the identifier, and conditional counting and summing functions produce totals for any category (Alexander, Kusleika and Walkenbach, 2018). For example, the SUMIFS function adds the values that meet more than one condition, such as the fees of one department that are still unpaid (Microsoft, n.d.a).

Excel also provides data validation, which restricts what can be typed into a cell, and conditional formatting, which changes the appearance of a cell according to its value. Together they help keep data consistent and make important cases visible without reading every row.

2.2 Dashboard Design

Few (2006) explains that a dashboard should gather the most important information on a single screen so that it can be monitored at a glance. Clear labels, restrained colour and simple chart types help the reader understand the figures quickly, while decoration that does not carry information distracts from them. These ideas guided the layout of the ClubSphere dashboard: key numbers appear as cards at the top, alerts follow, and charts are kept simple and consistently coloured.

2.3 Problem Statement

The coordinator of a college club needs quick answers to the following questions: How many members are active, expired or about to expire? Who has paid the fee, who owes money and which payments are overdue? Which events were well attended and which departments take part most? How much money has been collected, and can the details of any member be found instantly? When records are scattered in unrelated lists, these answers require repeated manual work and are prone to error.

2.4 Gap Identification

In the author's observation, small clubs usually keep either a plain member list or several unconnected sheets. A plain list stores data but does not calculate anything, while separate sheets must be compared by hand. Dedicated membership software could solve this but may be costly or complex for a small student club. Table 2.1 compares the manual approach with the proposed workbook.

Table 2.1: Comparison of manual records and ClubSphere

Aspect	Manual / separate lists	ClubSphere workbook
Membership status	Checked by reading dates	Calculated automatically as Active or Expired
Fee balance	Calculated by hand	Amount due minus amount paid, with Paid / Overdue status
Attendance	Counted manually per event	Counted by formula for each event and member
Member details	Searched in several lists	One-page profile from a Member ID
Reports	Prepared separately	Formula-driven tables and charts that update
Errors	Typing mistakes go unnoticed	Dropdowns, validation rules and colour alerts

The gap addressed by this project is therefore a low-cost, no-code tool that connects membership, fee, event and attendance records and reports on them automatically.

2.5 Design Objectives

- Maintain a central member register with automatic renewal date and membership status.
- Track fees with automatic balance, payment status and days overdue.
- Record events and attendance and calculate attendance and seat-fill percentages.
- Provide a one-page Member Profile lookup using a Member ID.
- Generate department, fee, monthly and event reports using formulas and charts.
- Present key figures, alerts and charts on a single dashboard that updates automatically.
- Reduce data-entry errors using dropdown lists, validation rules and colour-coded statuses.

2.6 Design Constraints

- The project must be completed individually using Microsoft Excel (or Word) as required by the course.
- No macros, VBA or other programming may be used; only built-in features.

- The workbook must work offline on a single computer and use fictional sample data.
- Formulas should work in common Excel versions, so INDEX with MATCH is used instead of newer lookup functions.
- The workbook is sized for up to 100 members, 150 fee records, 20 events and 300 attendance records.

Chapter 3
UI Design Overview

3.1 Design Principles

The interface uses a navy and teal colour scheme with the Times-style simplicity of a business report: one font family, consistent alignment and gridlines hidden on presentation sheets. Input cells use blue text and formula cells use black text so that users know where to type. Status values are colour coded: green for good states (Active, Paid, Present), amber for warnings (Expiring Soon, Partial), red for problems (Expired, Overdue, Absent) and blue for pending states (Pending, Upcoming, Registered). Every data sheet has frozen headers, filters and dropdown lists.

3.2 Dashboard

The Dashboard is the first sheet. It shows eight key performance indicator cards (total, active, expired and expiring members, fees collected, outstanding dues, collection rate and average attendance rate), three automatically written alerts, six charts and two leaderboards listing the five most engaged members and the five largest pending dues (Figure 3.1).

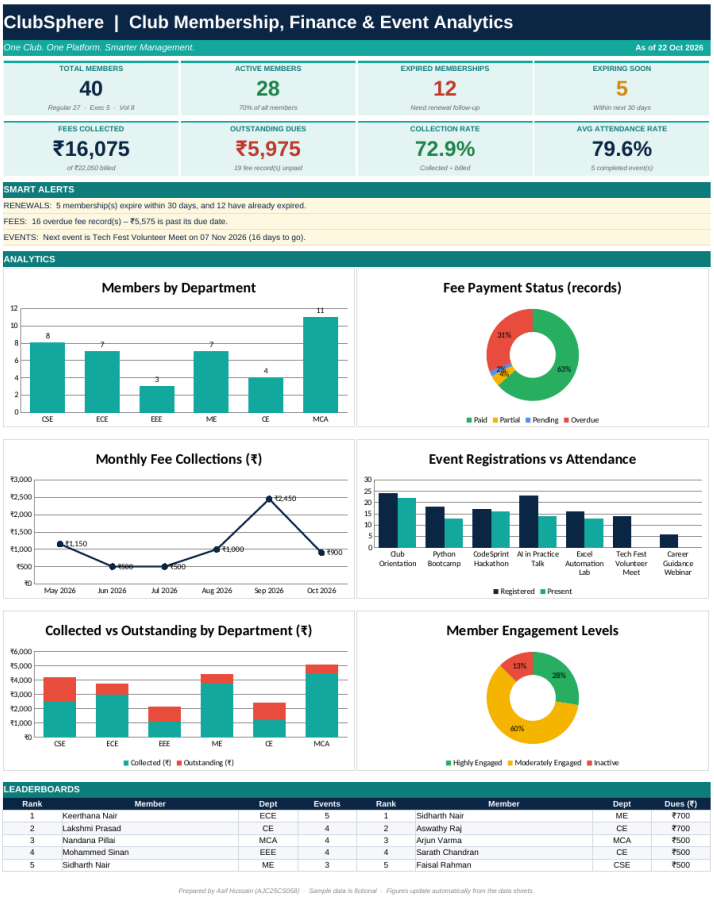


Figure 3.1: ClubSphere Dashboard

3.3 Member Register

The Member Register is the master record of the club. Blue columns are typed by the user, while the renewal date, membership status, days to renewal, renewal alert, events attended and fees outstanding are calculated. Expired and expiring memberships are highlighted, and a data bar shows the outstanding amount (Figure 3.2).

MEMBER REGISTER

Blue text = type / pick here · Black text = automatic formula (do not overwrite) · Sample data is fictional

Master record of every member. Member ID is the key that links Fees, Attendance and Profile.

Member ID	Full Name	Department	Year	Membership Type	Email	Phone	Joining Date	Renewal Date	Membership Status	Days to Renewal	Renewal Alert	Events Attended	Fees Outstanding (₹)
CS001	Lakshmi Prasad	CE	3	Regular	lakshmi.prasad@example.com	9586319898	27-Jun-2025	27-Jun-2026	Expired	-117	Expired	4	₹0
CS002	Rohan Thomas	MCA	1	Regular	rohan.thomas@example.com	9557174076	01-Jul-2025	01-Jul-2026	Expired	-113	Expired	2	₹0
CS003	Joel Abraham	MCA	2	Volunteer	joel.abraham@example.com	9169036396	28-Jul-2025	28-Jul-2026	Expired	-86	Expired	1	₹0
CS004	Leona Shaji	ME	4	Executive	leona.shaji@example.com	9497422744	01-Aug-2025	01-Aug-2026	Expired	-82	Expired	2	₹0
CS005	Fathima Nazar	CSE	4	Volunteer	fathima.nazar@example.com	9257151433	21-Aug-2025	21-Aug-2026	Expired	-62	Expired	2	₹150
CS006	Ashwin Kumar	ECE	1	Executive	ashwin.kumar@example.com	9198965313	16-Sep-2025	16-Sep-2026	Expired	-36	Expired	2	₹0
CS007	Basil Varghese	CSE	1	Regular	basil.varghese@example.com	9755126092	21-Sep-2025	21-Sep-2026	Expired	-31	Expired	1	₹250
CS008	Arjun Varma	MCA	1	Regular	arjun.varma@example.com	9229728319	04-Oct-2025	04-Oct-2026	Expired	-18	Expired	1	₹500
CS009	Sidharth Nair	ME	3	Regular	sidharth.nair@example.com	9817771848	07-Oct-2025	07-Oct-2026	Expired	-15	Expired	3	₹700
CS010	Sarath Chandran	CE	3	Regular	sarath.chandran@example.com	9266173260	10-Oct-2025	10-Oct-2026	Expired	-12	Expired	2	₹500
CS011	Aleena Jose	CSE	4	Volunteer	aleena.jose@example.com	9840620525	19-Oct-2025	19-Oct-2026	Expired	-3	Expired	3	₹150
CS012	Adithya Krishnan	CE	2	Regular	adithya.krishnan@example.com	9686052164	21-Oct-2025	21-Oct-2026	Expired	-1	Expired	2	₹0
CS013	Tom Jacob	CSE	2	Regular	tom.jacob@example.com	9742244082	23-Oct-2025	23-Oct-2026	Active	1	Expiring Soon	1	₹0

Figure 3.2: Member Register sheet

3.4 Fee Tracker

The Fee Tracker keeps one row for each fee. The member name, department and membership type are retrieved from the Member ID. The balance, payment status and days overdue are calculated, and the status cell turns green, amber, red or blue (Figure 3.3).

FEE TRACKER

Blue text = type / pick here · Black text = automatic formula (do not overwrite) · Sample data is fictional

One row per fee obligation. Amount Paid is the cumulative amount received. Balance and Payment Status calculate automatically.

Fee ID	Member ID	Member Name	Department	Membership Type	Fee Type	Amount Due (₹)	Amount Paid (₹)	Due Date	Payment Date	Balance (₹)	Payment Status	Days Overdue
FEE001	CS001	Lakshmi Prasad	CE	Regular	Membership Fee	₹500	₹500	12-Jul-2025	28-Jun-2025	₹0	Paid	0
FEE002	CS002	Rohan Thomas	MCA	Regular	Membership Fee	₹500	₹500	16-Jul-2025	10-Jul-2025	₹0	Paid	0
FEE003	CS003	Joel Abraham	MCA	Volunteer	Membership Fee	₹300	₹300	12-Aug-2025	03-Aug-2025	₹0	Paid	0
FEE004	CS004	Leona Shaji	ME	Executive	Membership Fee	₹750	₹750	16-Aug-2025	09-Aug-2025	₹0	Paid	0
FEE005	CS005	Fathima Nazar	CSE	Volunteer	Membership Fee	₹300	₹150	05-Sep-2025	06-Sep-2025	₹150	Overdue	412
FEE006	CS006	Ashwin Kumar	ECE	Executive	Membership Fee	₹750	₹750	01-Oct-2025	26-Sep-2025	₹0	Paid	0
FEE007	CS007	Basil Varghese	CSE	Regular	Membership Fee	₹500	₹250	06-Oct-2025	09-Oct-2025	₹250	Overdue	381
FEE008	CS008	Arjun Varma	MCA	Regular	Membership Fee	₹500	₹0	19-Oct-2025	Not paid	₹500	Overdue	368
FEE009	CS009	Sidharth Nair	ME	Regular	Membership Fee	₹500	₹0	22-Oct-2025	Not paid	₹500	Overdue	365
FEE010	CS010	Sarath Chandran	CE	Regular	Membership Fee	₹500	₹0	25-Oct-2025	Not paid	₹500	Overdue	362
FEE011	CS011	Aleena Jose	CSE	Volunteer	Membership Fee	₹300	₹150	03-Nov-2025	08-Nov-2025	₹150	Overdue	353
FEE012	CS012	Adithya Krishnan	CE	Regular	Membership Fee	₹500	₹500	05-Nov-2025	27-Oct-2025	₹0	Paid	0

Figure 3.3: Fee Tracker sheet

3.5 Event Manager

The Event Manager lists each event with its category, date, venue and capacity. Registered, present and absent counts, attendance percentage, seats filled percentage and event status are calculated from the Attendance sheet (Figure 3.4).

EVENT MANAGER

Blue text = type / pick here · Black text = automatic formula (do not overwrite) · Sample data is fictional

One row per event. Registered, Present and percentages are calculated from the Attendance sheet.

Event ID	Event Name	Category	Event Date	Venue	Capacity	Registered	Present	Absent	Attendance %	Seats Filled %	Status	Days to Event
EV001	Club Orientation	Orientation	08-Aug-2026	Seminar Hall	60	24	22	2	92%	40%	Completed	-75
EV002	Python Bootcamp	Workshop	29-Aug-2026	Computer Lab 1	40	18	13	5	72%	45%	Completed	-54
EV003	CodeSprint Hackathon	Competition	19-Sep-2026	Main Auditorium	50	17	16	1	94%	34%	Completed	-33
EV004	AI in Practice Talk	Talk	03-Oct-2026	Seminar Hall	80	23	14	9	61%	29%	Completed	-19
EV005	Excel Automation Lab	Workshop	17-Oct-2026	Computer Lab 2	40	16	13	3	81%	40%	Completed	-5
EV006	Tech Fest Volunteer Meet	Meeting	07-Nov-2026	Room 204	60	14	0	0	N/A	23%	Upcoming	16
EV007	Career Guidance Webinar	Webinar	21-Nov-2026	Online	100	6	0	0	N/A	6%	Upcoming	30

Figure 3.4: Event Manager sheet

3.6 Attendance

The Attendance sheet contains one row for every member-event registration. The user selects the Event ID and Member ID from dropdown lists and sets the status to Present, Absent or Registered. The member name, department and event name are filled automatically (Figure 3.5).

ATTENDANCE

Blue text = type / pick here · Black text = automatic formula (do not overwrite) · Sample data is fictional

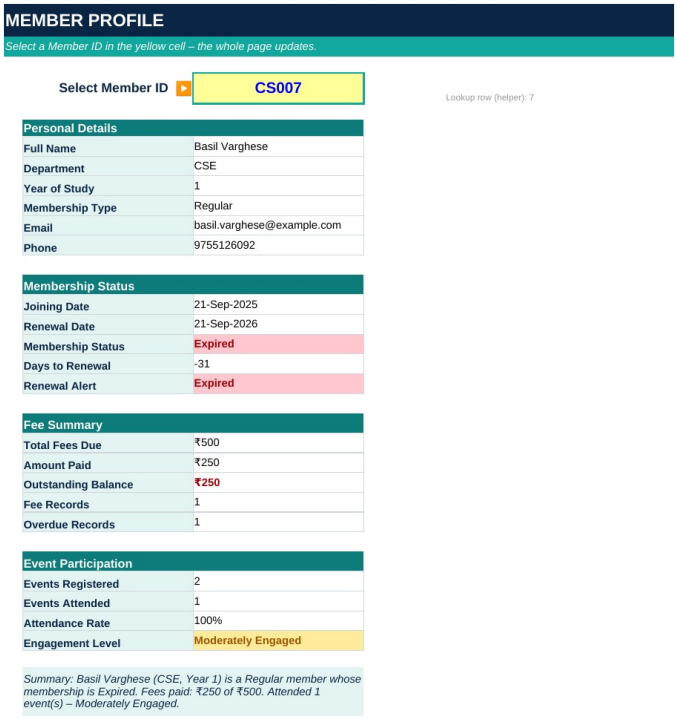
One row per member-event registration. Never put several Member IDs in one cell.

Reg ID	Event ID	Member ID	Member Name	Department	Event Name	Attendance Status
REG001	EV001	CS001	Lakshmi Prasad	CE	Club Orientation	Present
REG002	EV001	CS003	Joel Abraham	MCA	Club Orientation	Present
REG003	EV001	CS004	Leona Shaji	ME	Club Orientation	Present
REG004	EV001	CS006	Ashwin Kumar	ECE	Club Orientation	Present
REG005	EV001	CS007	Basil Varghese	CSE	Club Orientation	Present
REG006	EV001	CS008	Arjun Varma	MCA	Club Orientation	Absent
REG007	EV001	CS009	Sidharth Nair	ME	Club Orientation	Present
REG008	EV001	CS011	Aleena Jose	CSE	Club Orientation	Present
REG009	EV001	CS012	Adithya Krishnan	CE	Club Orientation	Present
REG010	EV001	CS014	Nandana Pillai	MCA	Club Orientation	Present
REG011	EV001	CS015	Devika Suresh	ECE	Club Orientation	Present
REG012	EV001	CS016	Aarav Nair	MCA	Club Orientation	Present
REG013	EV001	CS017	Keerthana Nair	ECE	Club Orientation	Present
REG014	EV001	CS020	Faisal Rahman	CSE	Club Orientation	Absent
REG015	EV001	CS021	Nikhil Mathew	ECE	Club Orientation	Present
REG016	EV001	CS026	Vishnu Mohan	ME	Club Orientation	Present

Figure 3.5: Attendance sheet

3.7 Member Profile

The Member Profile sheet has a yellow selector cell. When a Member ID is chosen, the page displays personal details, membership status, fee summary, event participation, an engagement level and a one-line summary sentence. It can be printed as a one-page member summary (Figure 3.6).



MEMBER PROFILE

Select a Member ID in the yellow cell – the whole page updates.

Select Member ID: CS007

Lookup row (helper): 7

Personal Details	
Full Name	Basil Varghese
Department	CSE
Year of Study	1
Membership Type	Regular
Email	basil.varghese@example.com
Phone	9755126092

Membership Status	
Joining Date	21-Sep-2025
Renewal Date	21-Sep-2026
Membership Status	Expired
Days to Renewal	-31
Renewal Alert	Expired

Fee Summary	
Total Fees Due	₹500
Amount Paid	₹250
Outstanding Balance	₹250
Fee Records	1
Overdue Records	1

Event Participation	
Events Registered	2
Events Attended	1
Attendance Rate	100%
Engagement Level	Moderately Engaged

Summary: Basil Varghese (CSE, Year 1) is a Regular member whose membership is Expired. Fees paid: ₹250 of ₹500. Attended 1 event(s) – Moderately Engaged.

Figure 3.6: Member Profile sheet

3.8 Reports

The Reports sheet contains six formula-driven tables: department-wise summary, membership-type summary, payment-status summary, monthly collections for the last six months, event performance and member engagement levels. The charts on the dashboard read their data from these tables (Figure 3.7).

REPORTS

All tables are formula-driven and feed the Dashboard charts.

1. Department-wise Summary

Department	Members	Active	Expired	Fees Due (₹)	Collected (₹)	Outstanding (₹)	Events Attended	Avg Events / Member
CSE	8	5	3	₹4,200	₹2,550	₹1,650	13	1.63
ECE	7	6	1	₹3,750	₹3,000	₹750	17	2.43
EEE	3	3	0	₹2,150	₹1,075	₹1,075	6	2.00
ME	7	5	2	₹4,450	₹3,750	₹700	12	1.71
CE	4	1	3	₹2,400	₹1,200	₹1,200	10	2.50
MCA	11	8	3	₹5,100	₹4,500	₹600	20	1.82
Total	**40**	**28**	**12**	**₹22,050**	**₹16,075**	**₹5,975**	**78**	**1.95**

2. Membership Type Summary

Membership Type	Members	Active	Fees Due (₹)	Collected (₹)	Outstanding (₹)
Regular	27	20	₹15,500	₹10,300	₹5,200
Executive	5	3	₹3,750	₹3,375	₹375
Volunteer	8	5	₹2,800	₹2,400	₹400
Total	**40**	**28**	**₹22,050**	**₹16,075**	**₹5,975**

3. Payment Status Summary

Payment Status	Fee Records	Amount Due (₹)	Collected (₹)	Outstanding (₹)
Paid	33	₹14,200	₹14,200	₹0
Partial	2	₹400	₹200	₹200
Pending	1	₹200	₹0	₹200
Overdue	16	₹7,250	₹1,675	₹5,575
Total	**52**	**₹22,050**	**₹16,075**	**₹5,975**

4. Monthly Collections (last 6 months)

Month	Month Start	Collected (₹)	Payments Received
May 2026	01-May-2026	₹1,150	4
Jun 2026	01-Jun-2026	₹500	1
Jul 2026	01-Jul-2026	₹500	1
Aug 2026	01-Aug-2026	₹1,000	2
Sep 2026	01-Sep-2026	₹2,450	9
Oct 2026	01-Oct-2026	₹900	4

5. Event Performance

Event	Date	Capacity	Registered	Present	Attendance %	Seats Filled %
Club Orientation	08-Aug-2026	60	24	22	92%	40%
Python Bootcamp	29-Aug-2026	40	18	13	72%	45%
CodeSprint Hackathon	19-Sep-2026	50	17	16	94%	34%
AI in Practice Talk	03-Oct-2026	80	23	14	61%	29%
Excel Automation Lab	17-Oct-2026	40	16	13	81%	40%
Tech Fest Volunteer Meet	07-Nov-2026	60	14	0	N/A	23%
Career Guidance Webinar	21-Nov-2026	100	6	0	N/A	6%

Shows the first 7 events from the Event Manager. Attendance % appears only after an event is completed.

6. Member Engagement Levels

Engagement Level	Members	% of Members	Rule
Highly Engaged	11	28%	3 or more events attended
Moderately Engaged	24	60%	1 to 2 events attended
Inactive	5	13%	No events attended

Figure 3.7: Reports sheet

Chapter 4
Implementation

4.1 Workbook Structure

The workbook ClubSphere_Working_Prototype.xlsx contains nine sheets, listed in Table 4.1. The sample data consists of 40 members from six departments (CSE, ECE, EEE, ME, CE and MCA), 52 fee records, 7 events and 118 attendance records. All names, contact details, amounts and dates are fictional. Unused spare rows and helper columns are hidden so that the sheets show only complete records; the hidden rows can be unhidden to add new members, fees, events or attendance.

Table 4.1: Sheets of the ClubSphere workbook

Sheet	Purpose
Dashboard	KPI cards, smart alerts, six charts and Top 5 leaderboards
Member Register	Master member record with calculated renewal date, status, alert, events attended and dues
Fee Tracker	Fee records with calculated balance, payment status and days overdue
Event Manager	Event details with calculated registrations, attendance % and status
Attendance	Member-event registrations with Present / Absent / Registered status
Member Profile	Dropdown-driven one-page lookup of a selected member
Reports	Six summary tables that feed the dashboard charts
Read Me	Usage guide, colour code, formulas and assumptions
Lists	Dropdown lists and settings (report date, alert window, validity)

4.2 Data Design and Relationships

Each record has a unique identifier. Member ID links the Member Register, Fee Tracker and Attendance sheets, and Event ID links the Event Manager and Attendance sheets. Names and departments are never typed twice; they are retrieved with INDEX and MATCH. The Reports sheet summarises the four data sheets and the Dashboard reads from both (Figure 4.1).

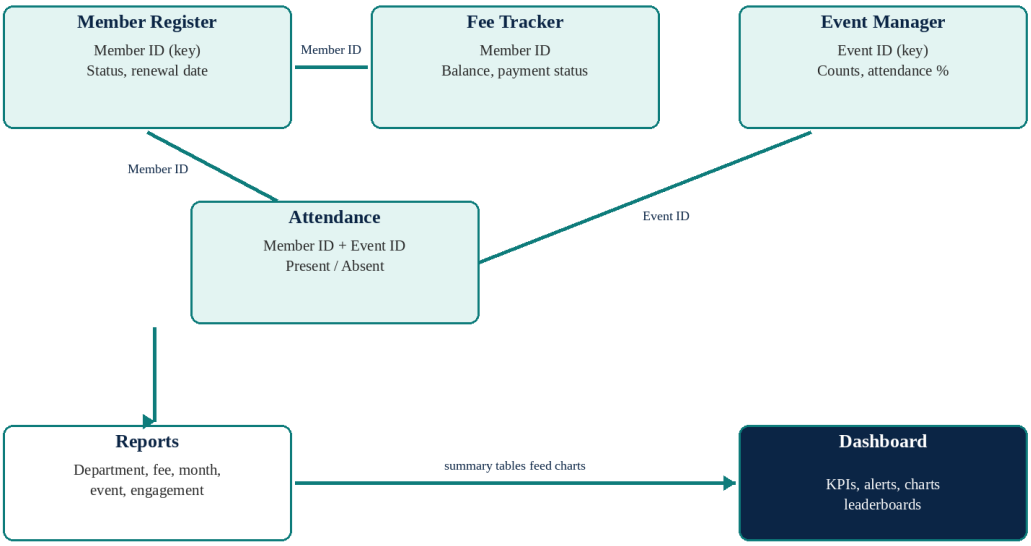


Figure 4.1: Relationship between the sheets of ClubSphere

Three settings on the Lists sheet control the date logic: the Report Date (fixed at 22 October 2026 so that results match this report; =TODAY() can be typed there for live alerts), the Renewal Alert Window (30 days) and the Membership Validity (12 months).

4.3 Steps Followed to Develop the Project

Step 1 – Create the workbook. The nine sheets were created and named, a navy and teal colour theme was chosen and sheet tabs were coloured.

Step 2 – Lists and settings. The dropdown values for departments, membership types, attendance statuses, fee types, years and event categories were entered on the Lists sheet together with the three settings. They were named ReportDate, AlertDays and ValidMonths.

Step 3 – Master data sheets. Headers were placed in row 4 and data from row 5 on the four data sheets. Fictional records were entered, and frozen panes, filters, borders and number formats were applied (dates as dd-mmm-yyyy, amounts in rupees).

Step 4 – Connect the records. Member ID and Event ID were used consistently, and INDEX with MATCH was used to retrieve names, departments and event names.

Step 5 – Add formulas. Formulas were added for renewal date, membership status, balance, payment status, event counts and percentages (Table 4.2).

Step 6 – Validation and alerts. Dropdown lists, unique-ID, date and amount rules and conditional formatting were configured (Table 4.3).

Step 7 – Reports. The six summary tables were built with COUNTIFS and SUMIFS, including a six-month collection table that moves with the report date.

Step 8 – Dashboard. Eight KPI cards, three alert sentences, six charts and two leaderboards (LARGE with MATCH) were created and linked to the report tables.

Step 9 – Testing and documentation. The workbook was tested with changed data, screenshots were captured and the report and video were prepared.

4.4 Key Formulas

Table 4.2 lists the main formulas for the first data row of each sheet; they are copied down for all rows. In the workbook the ranges are fixed ranges such as 'Member Register'! $A$5:$A$104; readable names are shown here for clarity.

Table 4.2: Key formulas used in ClubSphere

Result	Formula	Sheet
Renewal Date	`=IF(H5="","",EDATE(H5,ValidMonths))`	Member Register
Membership Status	`=IF(I5<ReportDate,"Expired","Active")  (with blank and Not Set checks)`	Member Register
Renewal Alert	`=IF(K5<0,"Expired",IF(K5<=AlertDays,"Expiring Soon","OK"))`	Member Register
Events Attended	`=COUNTIFS(AttMemberID, $A5,AttStatus,"Present")`	Member Register
Name lookup	`=IFERROR(INDEX(MemberNames,MATCH($B5,MemberIDs,0)),"ID not found")`	Fee Tracker
Balance	`=G5-H5`	Fee Tracker
Payment Status	`=IF(K5<=0,"Paid",IF(I5<ReportDate,"Overdue",IF(H5>0,"Partial","Pending")))`	Fee Tracker
Registered	`=COUNTIFS(AttEventID,$A5)`	Event Manager
Attendance %	`=IF(L5<>"Completed","",H5/G5)`	Event Manager
Department fees	`=SUMIFS(AmountPaid,FeeDepartment,B5)`	Reports
Monthly collection	`=SUMIFS(AmountPaid,PaymentDate,">="&C30,PaymentDate,"<"&EDATE(C30,1))`	Reports
Top 5 engaged	`=INDEX(Names,MATCH(LARGE(EngagementKey,k),EngagementKey,0))`	Dashboard

4.5 Data Validation and Conditional Formatting

Table 4.3: Validation rules and conditional formatting

Sheet / field	Rule	Purpose
Member ID (Register)	Must be unique: COUNTIF($A$5:$A$104,A5)=1	Prevent duplicate members
Department, Type, Year	Dropdown from the Lists sheet	Consistent categories
Joining Date, Event Date	Date after 01-Jan-2020	Reject invalid dates
Member ID (Fees, Attendance, Profile)	Dropdown from the Member Register	Prevent wrong IDs
Amount Paid	Decimal, not more than Amount Due	Prevent over-payment entries
Attendance Status	Dropdown: Present, Absent, Registered	Consistent statuses
Status columns	Green / amber / red / blue fills	Show problems at a glance
Outstanding amounts	Orange data bars	Compare dues visually

4.6 Development Screenshots

The following screenshots show the working parts of the workbook. Figure 4.2 shows the Lists sheet, which holds the dropdown values and the three workbook settings together with the student name and roll number used in the Dashboard footer. Figures 4.3 to 4.5 show the formulas used in the first data row of the Member Register, Fee Tracker and Event Manager, displayed as text. Figure 4.6 shows the Read Me sheet that explains the workbook to a new user.

Department	Membership Type	Attendance Status	Fee Type	Year	Event Category
CSE	Regular	Present	Membership Fee	1	Orientation
ECE	Executive	Absent	Workshop Fee	2	Workshop
EEE	Volunteer	Registered	Event Fee	3	Competition
ME			Late Fee	4	Talk
CE					Meeting
MCA					Webinar

Setting	Value
Report Date	22-Oct-2026
Renewal Alert Window (days)	30
Membership Validity (months)	12
Student Name	Your Name
Roll Number	Your Roll No

Edit these lists to change dropdown choices everywhere. Yellow cells in column J are workbook settings.

Figure 4.2: Lists sheet with dropdown values and settings

Member Register – formulas in row 5

Cell	Column heading	Formula (first data row)
I5	Renewal Date	`=IF(H5="","",EDATE(H5,ValidMonths))`
J5	Membership Status	`=IF($A5="","",IF(I5="","Not Set",IF(I5<ReportDate,"Expired","Active")))`
K5	Days to Renewal	`=IF(I5="","",I5-ReportDate)`
L5	Renewal Alert	`=IF(K5="","",IF(K5<0,"Expired",IF(K5<=AlertDays,"Expiring Soon","OK")))`
M5	Events Attended	`=IF($A5="","",COUNTIFS(Attendance!$C$5:$C$304,$A5,Attendance!$G$5:$G$304,"Present"))`
N5	Fees Outstanding (₹)	`=IF($A5="","",SUMIFS('Fee Tracker'!$K$5:$K$154,'Fee Tracker'!$B$5:$B$154,$A5))`

Figure 4.3: Formulas in the Member Register (formula view)

Fee Tracker – formulas in row 5		
Cell	**Column heading**	**Formula (first data row)**
C5	Member Name	`=IF($B5="","",IFERROR(INDEX('Member Register'!$B$5:$B$104,MATCH($B5,'Member Register'!$A$5:$A$104,0)),"ID not found"))`
D5	Department	`=IF($B5="","",IFERROR(INDEX('Member Register'!$C$5:$C$104,MATCH($B5,'Member Register'!$A$5:$A$104,0)),"ID not found"))`
E5	Membership Type	`=IF($B5="","",IFERROR(INDEX('Member Register'!$E$5:$E$104,MATCH($B5,'Member Register'!$A$5:$A$104,0)),"ID not found"))`
K5	Balance (₹)	`=IF(G5="","",G5-H5)`
L5	Payment Status	`=IF(G5="","",IF(K5<=0,"Paid",IF(I5<ReportDate,"Overdue",IF(H5>0,"Partial","Pending"))))`
M5	Days Overdue	`=IF(L5="","",IF(L5="Overdue",ReportDate-I5,0))`

Figure 4.4: Formulas in the Fee Tracker (formula view)

Event Manager – formulas in row 5		
Cell	**Column heading**	**Formula (first data row)**
G5	Registered	`=IF($A5="","",COUNTIFS(Attendance!$B$5:$B$304,$A5))`
H5	Present	`=IF($A5="","",COUNTIFS(Attendance!$B$5:$B$304,$A5,Attendance!$G$5:$G$304,"Present"))`
I5	Absent	`=IF($A5="","",COUNTIFS(Attendance!$B$5:$B$304,$A5,Attendance!$G$5:$G$304,"Absent"))`
J5	Attendance %	`=IF($A5="","",IF(L5<>"Completed","N/A",IF(G5=0,0,H5/G5)))`
K5	Seats Filled %	`=IF($A5="","",IF(F5=0,0,G5/F5))`
L5	Status	`=IF($A5="","",IF(D5<ReportDate,"Completed",IF(D5=ReportDate,"Today","Upcoming")))`
M5	Days to Event	`=IF($A5="","",D5-ReportDate)`

Figure 4.5: Formulas in the Event Manager (formula view)

READ ME – HOW TO USE CLUBSPHERE

Course 24MNC201 · Essentials of Office Automation · Individual Mini-Project

PURPOSE	
Problem	Club records (members, fees, events, attendance) are often kept in separate lists, which makes renewals, dues and reports slow and error-prone.
Solution	ClubSphere links all records through Member ID / Event ID and uses formulas to calculate statuses, balances and analytics automatically.
SHEET GUIDE	
Dashboard	KPI cards, smart alerts, 6 charts and leaderboards.
Member Register	Master member list; renewal date, status and alerts are calculated.
Fee Tracker	Fee records with automatic balance, payment status and days overdue.
Event Manager	Event list with automatic registration counts, attendance % and status.
Attendance	One row per member-event registration (Present / Absent / Registered).
Member Profile	Pick a Member ID to see a one-page summary of that member.
Reports	Department, type, payment, monthly, event and engagement summaries.
Lists	Dropdown lists and settings (Report Date, alert window, membership validity).
HOW TO ADD DATA	
New member	Type the next free row in Member Register: ID, name, department (dropdown), year, type, email, phone, joining date. Everything else fills itself.
New fee	Add a row in Fee Tracker: pick Member ID, fee type, amount due, amount paid, due date, payment date.
New event	Add a row in Event Manager. Then add registrations in Attendance.
Capacity	Member Register: 100 members · Fee Tracker: 150 records · Event Manager: 20 events · Attendance: 300 rows. Unused rows are hidden: select the rows around the gap, right-click and choose Unhide to add records.
COLOUR CODE	
Blue text	Input – type or pick it yourself.
Black text	Formula – calculated automatically; do not overwrite.
Yellow cell	Selector / setting you are expected to change (Member Profile, Lists).
Green / amber / red / blue fills	Status highlights from conditional formatting (Active/Paid/Present · Expiring/Partial · Expired/Overdue/Absent · Pending/Upcoming/Registered).
KEY FORMULAS	
Renewal Date	Joining Date + ValidMonths, using EDATE(Joining Date, ValidMonths)
Membership Status	IF(Renewal Date < Report Date, Expired, Active), with a Not Set check
Balance	Amount Due minus Amount Paid
Payment Status	Paid if balance ≤ 0; Overdue if due date passed; Partial if some paid; otherwise Pending.
Lookups	INDEX + MATCH on Member ID / Event ID.
Counts & sums	COUNTIFS, SUMIFS, SUMPRODUCT-free summaries; LARGE + MATCH for leaderboards.
ASSUMPTIONS	
Sample data	All names, emails, phone numbers, amounts and dates are fictional, generated for demonstration.
Fee schedule	Membership fee: Regular ₹500, Executive ₹750, Volunteer ₹300. Workshop fee ₹200. Typed as inputs in Fee Tracker.
Membership validity	12 months from joining date (editable in Lists!J4).
Report Date	Fixed at 22-Oct-2026 (Lists!J2) so every figure matches the report and slides. Type =TODAY() there for live alerts.
Your details	Enter your name and roll number in Lists!J5 and J6; they appear in the Dashboard footer.
Helper columns	Hidden columns O:P in Member Register and N in Event Manager support the leaderboards and next-event alert (unhide to inspect).
No code	No macros/VBA – only built-in Excel features: tables, formulas, validation, conditional formatting and charts.

Figure 4.6: Read Me sheet of the workbook

Chapter 5
Testing and Results

5.1 Testing

The workbook was tested by changing data in a copy, recalculating and comparing the output with the original. Each test behaved as expected (Table 5.1).

Table 5.1: Test cases and results

No.	Test action	Expected result	Actual result	Status
1	Change Amount Paid of fee FEE005 (member CS005) from Rs. 150 to Rs. 300	Status becomes Paid; collected rises and outstanding falls by Rs. 150	Collected Rs. 16,075 to Rs. 16,225; outstanding Rs. 5,975 to Rs. 5,825	Pass
2	Add new member CS041 (CSE, Regular, joined 20-Oct-2026)	Renewal date 20-Oct-2027, status Active; total and active members increase by 1	Renewal 20-Oct-2027, Active; total 40 to 41; active 28 to 29	Pass
3	Select Member ID CS012 on Member Profile	Profile shows details of CS012	Adithya Krishnan, CE, Expired, Rs. 700 paid, 2 events, outstanding Rs. 0	Pass
4	Change Report Date from 22-Oct-2026 to 01-Dec-2026	More memberships expire	Expired 12 to 18; active 28 to 22; expiring soon 5 to 1	Pass

The screenshots below were captured from the workbook before and after each change. In Test 1 the Amount Paid of fee FEE005 was changed from Rs. 150 to Rs. 300; the Fee Tracker row changed from Overdue to Paid (Figure 5.1) and the Dashboard totals changed accordingly (Figure 5.2).

Before: FEE005 balance Rs. 150, status Overdue

FEE TRACKER

Blue text = type / pick here · Black text = automatic formula (do not overwrite) · Sample data is fictional

One row per fee obligation. Amount Paid is the cumulative amount received. Balance and Payment Status calculate automatically.

Fee ID	Member ID	Member Name	Department	Membership Type	Fee Type	Amount Due (₹)	Amount Paid (₹)	Due Date	Payment Date	Balance (₹)	Payment Status	Days Overdue
FEE001	CS001	Lakshmi Prasad	CE	Regular	Membership Fee	₹500	₹500	12-Jul-2025	28-Jun-2025	₹0	Paid	0
FEE002	CS002	Rohan Thomas	MCA	Regular	Membership Fee	₹500	₹500	16-Jul-2025	10-Jul-2025	₹0	Paid	0
FEE003	CS003	Joel Abraham	MCA	Volunteer	Membership Fee	₹300	₹300	12-Aug-2025	03-Aug-2025	₹0	Paid	0
FEE004	CS004	Leona Shaji	ME	Executive	Membership Fee	₹750	₹750	16-Aug-2025	09-Aug-2025	₹0	Paid	0
FEE005	CS005	Fathima Nazar	CSE	Volunteer	Membership Fee	₹300	₹150	05-Sep-2025	06-Sep-2025	₹150	Overdue	412
FEE006	CS006	Ashwin Kumar	ECE	Executive	Membership Fee	₹750	₹750	01-Oct-2025	26-Sep-2025	₹0	Paid	0
FEE007	CS007	Basil Varghese	CSE	Regular	Membership Fee	₹500	₹250	06-Oct-2025	09-Oct-2025	₹250	Overdue	381
FEE008	CS008	Arjun Varma	MCA	Regular	Membership Fee	₹500	₹0	19-Oct-2025	Not paid	₹500	Overdue	368
FEE009	CS009	Sidharth Nair	ME	Regular	Membership Fee	₹500	₹0	22-Oct-2025	Not paid	₹500	Overdue	365
FEE010	CS010	Sarath Chandran	CE	Regular	Membership Fee	₹500	₹0	25-Oct-2025	Not paid	₹500	Overdue	362
FEE011	CS011	Aleena Jose	CSE	Volunteer	Membership Fee	₹300	₹150	03-Nov-2025	08-Nov-2025	₹150	Overdue	353
FEE012	CS012	Adithya Krishnan	CE	Regular	Membership Fee	₹500	₹500	05-Nov-2025	27-Oct-2025	₹0	Paid	0

After: Amount Paid changed to Rs. 300, balance Rs. 0, status Paid

FEE TRACKER

Blue text = type / pick here · Black text = automatic formula (do not overwrite) · Sample data is fictional

One row per fee obligation. Amount Paid is the cumulative amount received. Balance and Payment Status calculate automatically.

Fee ID	Member ID	Member Name	Department	Membership Type	Fee Type	Amount Due (₹)	Amount Paid (₹)	Due Date	Payment Date	Balance (₹)	Payment Status	Days Overdue
FEE001	CS001	Lakshmi Prasad	CE	Regular	Membership Fee	₹500	₹500	12-Jul-2025	28-Jun-2025	₹0	Paid	0
FEE002	CS002	Rohan Thomas	MCA	Regular	Membership Fee	₹500	₹500	16-Jul-2025	10-Jul-2025	₹0	Paid	0
FEE003	CS003	Joel Abraham	MCA	Volunteer	Membership Fee	₹300	₹300	12-Aug-2025	03-Aug-2025	₹0	Paid	0
FEE004	CS004	Leona Shaji	ME	Executive	Membership Fee	₹750	₹750	16-Aug-2025	09-Aug-2025	₹0	Paid	0
FEE005	CS005	Fathima Nazar	CSE	Volunteer	Membership Fee	₹300	₹300	05-Sep-2025	06-Sep-2025	₹0	Paid	0
FEE006	CS006	Ashwin Kumar	ECE	Executive	Membership Fee	₹750	₹750	01-Oct-2025	26-Sep-2025	₹0	Paid	0
FEE007	CS007	Basil Varghese	CSE	Regular	Membership Fee	₹500	₹250	06-Oct-2025	09-Oct-2025	₹250	Overdue	381
FEE008	CS008	Arjun Varma	MCA	Regular	Membership Fee	₹500	₹0	19-Oct-2025	Not paid	₹500	Overdue	368
FEE009	CS009	Sidharth Nair	ME	Regular	Membership Fee	₹500	₹0	22-Oct-2025	Not paid	₹500	Overdue	365
FEE010	CS010	Sarath Chandran	CE	Regular	Membership Fee	₹500	₹0	25-Oct-2025	Not paid	₹500	Overdue	362
FEE011	CS011	Aleena Jose	CSE	Volunteer	Membership Fee	₹300	₹150	03-Nov-2025	08-Nov-2025	₹150	Overdue	353
FEE012	CS012	Adithya Krishnan	CE	Regular	Membership Fee	₹500	₹500	05-Nov-2025	27-Oct-2025	₹0	Paid	0

Figure 5.1: Test 1: Fee Tracker before and after paying FEE005

Before: fees collected Rs. 16,075 and outstanding Rs. 5,975

ClubSphere | Club Membership, Finance & Event Analytics

One Club. One Platform. Smarter Management. — As of 22 Oct 2026

TOTAL MEMBERS	ACTIVE MEMBERS	EXPIRED MEMBERSHIPS	EXPIRING SOON
40	28	12	5
Regular 27 · Exec 5 · Vol 8	70% of all members	Need renewal follow-up	Within next 30 days

FEES COLLECTED	OUTSTANDING DUES	COLLECTION RATE	AVG ATTENDANCE RATE
₹16,075	₹5,975	72.9%	79.6%
of ₹22,050 billed	19 fee record(s) unpaid	Collected ÷ billed	5 completed event(s)

After: fees collected Rs. 16,225 and outstanding Rs. 5,825

ClubSphere | Club Membership, Finance & Event Analytics

One Club. One Platform. Smarter Management. — As of 22 Oct 2026

TOTAL MEMBERS	ACTIVE MEMBERS	EXPIRED MEMBERSHIPS	EXPIRING SOON
40	28	12	5
Regular 27 · Exec 5 · Vol 8	70% of all members	Need renewal follow-up	Within next 30 days

FEES COLLECTED	OUTSTANDING DUES	COLLECTION RATE	AVG ATTENDANCE RATE
₹16,225	₹5,825	73.6%	79.6%
of ₹22,050 billed	18 fee record(s) unpaid	Collected ÷ billed	5 completed event(s)

Figure 5.2: Test 1: Dashboard totals before and after the payment

In Test 2 a new member CS041 was typed into the next row of the Member Register. The renewal date, status and days to renewal were filled in automatically (Figure 5.3).

MEMBER REGISTER

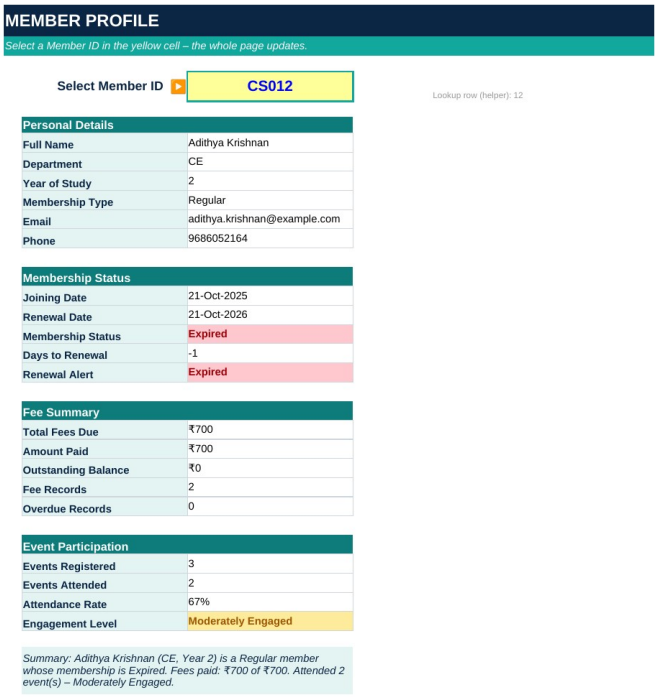
Blue text = type / pick here · Black text = automatic formula (do not overwrite) · Sample data is fictional

Master record of every member. Member ID is the key that links Fees, Attendance and Profile.

				Membership					Membership	Days to		Events	Fees
CS033	Aswathy Raj	CE	1	Regular	aswathy.raj@example.com	9238662776	10-Jul-2026	10-Jul-2027	Active	261	OK	2	₹700
CS034	Sneha Pillai	ME	2	Regular	sneha.pillai@example.com	9943326434	14-Jul-2026	14-Jul-2027	Active	265	OK	0	₹0
CS035	Diya Thomas	MCA	1	Regular	diya.thomas@example.com	9272218179	31-Jul-2026	31-Jul-2027	Active	282	OK	2	₹0
CS036	Gayathri Menon	ME	2	Regular	gayathri.menon@example.com	9518728644	11-Aug-2026	11-Aug-2027	Active	293	OK	2	₹0
CS037	Reshma Biju	CSE	4	Regular	reshma.biju@example.com	9184459259	24-Aug-2026	24-Aug-2027	Active	306	OK	2	₹250
CS038	Hridya Anil	ECE	2	Regular	hridya.anil@example.com	9469438031	03-Sep-2026	03-Sep-2027	Active	316	OK	1	₹0
CS039	Ananya Menon	CSE	4	Regular	ananya.menon@example.com	9634296521	13-Sep-2026	13-Sep-2027	Active	326	OK	0	₹0
CS040	Riya Mathew	MCA	2	Regular	riya.mathew@example.com	9408883592	14-Oct-2026	14-Oct-2027	Active	357	OK	1	₹0
CS041	Test Member	CSE	1	Regular	test.member@example.com	9000000000	20-Oct-2026	20-Oct-2027	Active	363	OK	0	₹0

Figure 5.3: Test 2: new member CS041 added to the Member Register

In Test 3 the Member ID CS012 was selected on the Member Profile sheet, and the whole page changed to show that member (Figure 5.4).



MEMBER PROFILE

Select a Member ID in the yellow cell – the whole page updates.

Select Member ID: CS012

Lookup row (helper): 12

Personal Details	
Full Name	Adithya Krishnan
Department	CE
Year of Study	2
Membership Type	Regular
Email	adithya.krishnan@example.com
Phone	9686052164

Membership Status	
Joining Date	21-Oct-2025
Renewal Date	21-Oct-2026
Membership Status	Expired
Days to Renewal	-1
Renewal Alert	Expired

Fee Summary	
Total Fees Due	₹700
Amount Paid	₹700
Outstanding Balance	₹0
Fee Records	2
Overdue Records	0

Event Participation	
Events Registered	3
Events Attended	2
Attendance Rate	67%
Engagement Level	Moderately Engaged

Summary: Adithya Krishnan (CE, Year 2) is a Regular member whose membership is Expired. Fees paid: ₹700 of ₹700. Attended 2 event(s) – Moderately Engaged.

Figure 5.4: Test 3: Member Profile for Member ID CS012

In Test 4 the report date in the Lists sheet was changed from 22 October 2026 to 1 December 2026, and the membership counts on the Dashboard moved from active to expired (Figure 5.5).

Report date 22 Oct 2026: 28 active, 12 expired, 5 expiring soon



Report date 01 Dec 2026: 22 active, 18 expired, 1 expiring soon



Figure 5.5: Test 4: Dashboard counts for two different report dates

5.2 Results

With the sample data and a report date of 22 October 2026, ClubSphere produced the indicators in Table 5.2.

Table 5.2: Key indicators on the dashboard

Indicator	Result
Total members	40
Active members	28 (70%)
Expired memberships	12
Expiring within 30 days	5
Fees billed	Rs. 22,050
Fees collected	Rs. 16,075
Outstanding dues	Rs. 5,975
Collection rate	72.9%
Average event attendance rate	79.6%

Table 5.3: Department-wise summary

Dept	Members	Active	Expired	Fees Due	Collected	Outstanding	Events Attended
CSE	8	5	3	Rs. 4,200	Rs. 2,550	Rs. 1,650	13
ECE	7	6	1	Rs. 3,750	Rs. 3,000	Rs. 750	17
EEE	3	3	0	Rs. 2,150	Rs. 1,075	Rs. 1,075	6
ME	7	5	2	Rs. 4,450	Rs. 3,750	Rs. 700	12
CE	4	1	3	Rs. 2,400	Rs. 1,200	Rs. 1,200	10
MCA	11	8	3	Rs. 5,100	Rs. 4,500	Rs. 600	20
Total	**40**	**28**	**12**	**Rs. 22,050**	**Rs. 16,075**	**Rs. 5,975**	**78**

Table 5.4: Payment status summary

Status	Records	Amount Due	Collected	Outstanding
Paid	33	Rs. 14,200	Rs. 14,200	Rs. 0
Partial	2	Rs. 400	Rs. 200	Rs. 200
Pending	1	Rs. 200	Rs. 0	Rs. 200
Overdue	16	Rs. 7,250	Rs. 1,675	Rs. 5,575
Total	**52**	**Rs. 22,050**	**Rs. 16,075**	**Rs. 5,975**

Table 5.5: Event performance

Event	Date	Capacity	Registered	Present	Attendance	Seats Filled
Club Orientation	Aug 2026	60	24	22	91.7%	40.0%
Python Bootcamp	Aug 2026	40	18	13	72.2%	45.0%
CodeSprint Hackathon	Sep 2026	50	17	16	94.1%	34.0%
AI in Practice Talk	Oct 2026	80	23	14	60.9%	28.7%
Excel Automation Lab	Oct 2026	40	16	13	81.3%	40.0%
Tech Fest Volunteer Meet	Nov 2026	60	14	0	Upcoming	23.3%
Career Guidance Webinar	Nov 2026	100	6	0	Upcoming	6.0%

5.3 Observations

- MCA has the most members (11) and the highest collection (Rs. 4,500); ECE has the highest average attendance per member (2.4).
- CE and CSE each have three expired memberships, and CE has only one active member out of four, so they need renewal follow-up.
- Overdue fees account for Rs. 5,575 of the Rs. 5,975 outstanding, so reminders should target overdue records first.
- The Hackathon and the Orientation had the best attendance (94% and 92%) and the AI in Practice Talk the lowest (61%).
- Eleven members (27.5%) are highly engaged, 24 (60%) are moderately engaged and 5 (12.5%) have attended no event.

These observations apply to the fictional sample data and show the type of insight the workbook produces.

Chapter 6
Extension: Word Mail Merge

6.1 Introduction and Objectives

The Excel workbook stores and analyses the club's records, but the coordinator must also communicate with members. Microsoft Word mail merge combines a main document that contains the fixed text and merge fields with a data source that supplies one record for each recipient, and Word then produces a personalised document for every record (Microsoft, n.d.b). Bucki (2018) describes the same steps of choosing the document type, selecting recipients, inserting fields, previewing and finishing the merge. This extension, named ClubSphere Connect, uses the ClubSphere membership data as the data source.

The objectives of the extension are to prepare a clean data source from the workbook, to design Word main documents with merge fields for a confirmation letter, a renewal reminder and a fee payment notice, to design an ID card sheet with eight cards per page, to merge the documents and save the results as Word and PDF files, and to check that every record produces exactly one correct document.

6.2 Data Source

An Excel worksheet is the most convenient data source for mail merge. The first row must contain unique column headings that become the field names, and each following row must be one record (Alexander, Kusleika and Walkenbach, 2018). The data was therefore copied from the Member Register and Fee Tracker into a separate file, ClubSphere_MailMerge_Data.xlsx, with three worksheets: Members_All (40 records), Renewal_Due (17 records) and Fee_Due (16 records). Dates were stored as text in the form 27 June 2025 and amounts with the rupee sign so that Word shows them correctly, and helper fields such as FirstName, RenewalNote and FeeNote were added (Figure 6.1, Table 6.1).

MemberID	FullName	FirstName	Department	MembershipType	JoiningDate	RenewalDate	MembershipStatus	FeeDue	Balance	PaymentStatus
CS001	Lakshmi Prasad	Lakshmi	CE	Regular	27 June 2025	27 June 2026	Expired	₹500	₹0	Paid in full
CS002	Rohan Thomas	Rohan	MCA	Regular	1 July 2025	1 July 2026	Expired	₹500	₹0	Paid in full
CS003	Joel Abraham	Joel	MCA	Volunteer	28 July 2025	28 July 2026	Expired	₹300	₹0	Paid in full
CS004	Leona Shaji	Leona	ME	Executive	1 August 2025	1 August 2026	Expired	₹750	₹0	Paid in full
CS005	Fathima Nazar	Fathima	CSE	Volunteer	21 August 2025	21 August 2026	Expired	₹300	₹150	Overdue
CS006	Ashwin Kumar	Ashwin	ECE	Executive	16 September 2025	16 September 2026	Expired	₹750	₹0	Paid in full
CS007	Basil Varghese	Basil	CSE	Regular	21 September 2025	21 September 2026	Expired	₹500	₹250	Overdue
CS008	Arjun Varma	Arjun	MCA	Regular	4 October 2025	4 October 2026	Expired	₹500	₹500	Overdue
CS009	Sidharth Nair	Sidharth	ME	Regular	7 October 2025	7 October 2026	Expired	₹700	₹700	Overdue
CS010	Sarath Chandran	Sarath	CE	Regular	10 October 2025	10 October 2026	Expired	₹500	₹500	Overdue
CS011	Aleena Jose	Aleena	CSE	Volunteer	19 October 2025	19 October 2026	Expired	₹500	₹150	Overdue

Figure 6.1: Data source worksheet Members_All (selected columns)

Table 6.1: Main merge fields in the data source

Field name	Meaning	Example
MemberID	Unique member identifier	CS001
FullName	Full name of the member	Lakshmi Prasad
FirstName	First name used in the greeting	Lakshmi
Department	Department of the member	CE
MembershipType	Regular, Executive or Volunteer	Regular
JoiningDate	Date of joining	27 June 2025
RenewalDate	Membership valid until	27 June 2026
RenewalNote	Sentence about expiry	expired on 27 June 2026 (117 days ago)
Balance	Fees still to pay	₹150
FeeNote	Remark about the dues	1 overdue fee record(s); the oldest was due on 5 September 2025

6.3 Main Documents and Merged Output

Four main documents were designed in Word with the same letterhead: the club name, the tagline and the college name above a teal rule, followed by a date, the address block, the subject line, the body text and a member details table. Merge fields appear inside « » symbols in the main document and are replaced by real values in the merged output. The letter date is a DATE field in the main documents and is fixed at 22 October 2026 in the merged output to match the report date of the data.

The membership confirmation letter is merged with all 40 members, the renewal reminder only with the 17 members whose membership has expired or will expire within 30 days, and the fee payment notice only with the 16 members who have a balance (Figures 6.2 to 6.4).

Main document with merge fields

ClubSphere Student Club
One Club. One Platform. Smarter Management.
Amal Jyothi College of Engineering (Autonomous), Kanjirapally

28 October 2026
Ref: CS/2026/«MemberID»

«FullName»
Department of «Department», Year «Year»
«Email»

Subject: Membership Confirmation Letter (ID «MemberID»)

Dear «FirstName»,

We are pleased to confirm that you joined the ClubSphere Student Club on «JoiningDate» as a «MembershipType» member of the club.

Your membership details are given below:

Member ID	«MemberID»
Name	«FullName»
Department and year	«Department», Year «Year»
Membership type	«MembershipType»
Joining date	«JoiningDate»
Valid until	«RenewalDate»
Membership status	«MembershipStatus»
Events attended	«EventsAttended»
Fees paid	«AmountPaid» of «FeeDue»

As a member you can take part in club workshops, talks, competitions and meetings, and you will receive updates on upcoming events at «Email». Please keep your Member ID for all club activities.

Thank you for being a part of ClubSphere. We look forward to your active participation.

Yours sincerely,

The Club Coordinator
ClubSphere Student Club

Merged output

ClubSphere Student Club
One Club. One Platform. Smarter Management.
Amal Jyothi College of Engineering (Autonomous), Kanjirapally

22 October 2026
Ref: CS/2026/CS001

Lakshmi Prasad
Department of CE, Year 3
lakshmi.prasad@example.com

Subject: Membership Confirmation Letter (ID CS001)

Dear Lakshmi,

We are pleased to confirm that you joined the ClubSphere Student Club on 27 June 2025 as a Regular member of the club.

Your membership details are given below:

Member ID	CS001
Name	Lakshmi Prasad
Department and year	CE, Year 3
Membership type	Regular
Joining date	27 June 2025
Valid until	27 June 2026
Membership status	Expired
Events attended	4
Fees paid	₹500 of ₹500

As a member you can take part in club workshops, talks, competitions and meetings, and you will receive updates on upcoming events at lakshmi.prasad@example.com. Please keep your Member ID for all club activities.

Thank you for being a part of ClubSphere. We look forward to your active participation.

Yours sincerely,

The Club Coordinator
ClubSphere Student Club

Figure 6.2: Membership confirmation letter: main document and merged output

Main document with merge fields

ClubSphere Student Club
One Club. One Platform. Smarter Management.
Amal Jyothi College of Engineering (Autonomous), Kanjirapally

28 October 2026
Ref: CS/2026/«MemberID»

«FullName»
Department of «Department», Year «Year»
«Email»

Subject: Membership Renewal Reminder (ID «MemberID»)

Dear «FirstName»,

Our records show that your ClubSphere membership «RenewalNote».

To continue enjoying member benefits, please renew your membership by paying the renewal fee to the Club Coordinator. Your membership details are:

Member ID	«MemberID»
Membership type	«MembershipType»
Joining date	«JoiningDate»
Renewal date	«RenewalDate»
Current status	«MembershipStatus»
Renewal fee	«RenewalFee»

If you have already renewed, please ignore this letter and accept our thanks.

We value your participation and hope to continue working with you.

Yours sincerely,

The Club Coordinator
ClubSphere Student Club

Merged output

ClubSphere Student Club
One Club. One Platform. Smarter Management.
Amal Jyothi College of Engineering (Autonomous), Kanjirapally

22 October 2026
Ref: CS/2026/CS001

Lakshmi Prasad
Department of CE, Year 3
lakshmi.prasad@example.com

Subject: Membership Renewal Reminder (ID CS001)

Dear Lakshmi,

Our records show that your ClubSphere membership expired on 27 June 2026 (117 days ago).

To continue enjoying member benefits, please renew your membership by paying the renewal fee to the Club Coordinator. Your membership details are:

Member ID	CS001
Membership type	Regular
Joining date	27 June 2025
Renewal date	27 June 2026
Current status	Expired
Renewal fee	₹500

If you have already renewed, please ignore this letter and accept our thanks.

We value your participation and hope to continue working with you.

Yours sincerely,

The Club Coordinator
ClubSphere Student Club

Figure 6.3: Renewal reminder: main document and merged output

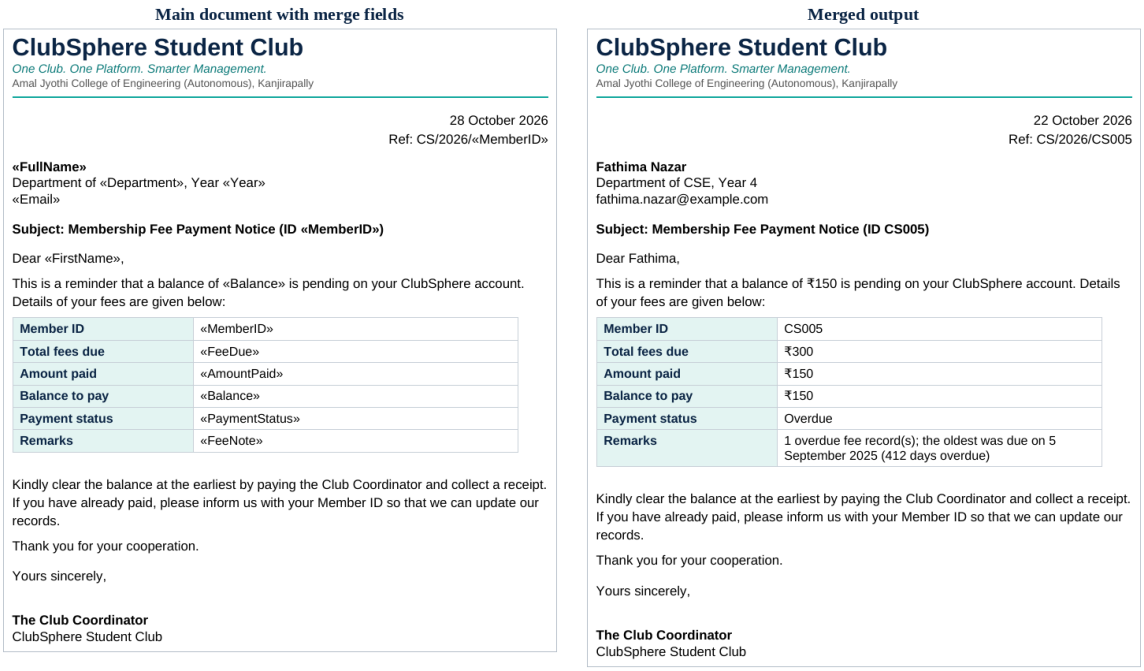


Figure 6.4: Fee payment notice: main document and merged output

The ID card sheet is a table of eight cards per page. The first card holds the merge fields and every other card begins with a Next Record field, so that eight different members appear on each page. The 40 members fill five pages (Figure 6.5).



Figure 6.5: Membership ID card sheet: main document and merged output (top half of page)

6.4 Steps Followed

The merge follows the five stages shown in Figure 6.6: start the merge by choosing Letters, select the Excel file and the correct worksheet as recipients, insert the merge fields, preview the results to check a few records, and finish by merging to a new document that is saved as a Word file and as a PDF.

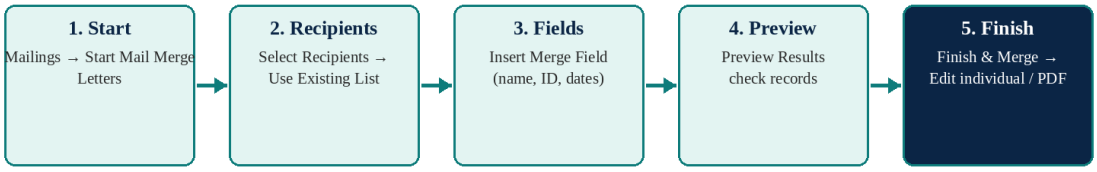


Figure 6.6: Mail merge workflow in Microsoft Word

The Word features used were merge fields, the DATE and NEXT fields, tables for the details block and the ID cards, paragraph borders and fonts for the letterhead, A4 page layout with page breaks between letters, and Save as PDF for sharing.

6.5 Testing and Results

The merged documents were checked against the data source (Table 6.2). Table 6.3 summarises the outputs.

Table 6.2: Test cases for the mail merge

No.	Test	Expected result	Actual result	Status
1	Confirmation letters for the 40 members in Members_All	40 letters, one per page	40 letters on 40 pages	Pass
2	Renewal reminders from Renewal_Due	17 reminders (12 expired and 5 expiring soon)	17 reminders on 17 pages	Pass
3	Fee notices from Fee_Due	16 notices, one per member with a balance	16 notices on 16 pages	Pass
4	Every member name appears in the merged output	All names present	No name missing in any file	Pass
5	No unreplaced field codes remain	No « » or field text in the output	None found	Pass
6	ID card layout	8 cards per page, 40 cards on 5 pages	5 pages with 8 cards each	Pass

Table 6.3: Merged outputs produced

Merged document	Data worksheet	Records	Pages	Formats
Membership confirmation letters	Members_All	40	40	Word, PDF
Renewal reminders	Renewal_Due	17	17	Word, PDF
Fee payment notices	Fee_Due	16	16	Word, PDF
Membership ID cards	Members_All	40	5	Word, PDF
Total		**113**	**78**	

6.6 Observations

- Four templates and one data file replaced 113 separately typed documents.
- Separate recipient lists meant that only the members who needed a renewal reminder or a fee notice received one.
- No names, dates or amounts had to be retyped, because every value came from the data source.
- Updating the data and merging again produces new documents in a few minutes.

Chapter 7
Conclusion and Future Scope

7.1 Conclusion

ClubSphere shows that Microsoft Excel can work as a small management system when records are linked through identifiers and formulas perform the calculations. Membership status, fee balances, attendance percentages, reports and dashboard figures update automatically when a record changes, and validation rules and colour coding reduce mistakes and make problems visible. The project met its objectives and demonstrates practical use of tables, formulas, functions, charts, formatting and page layout without any programming. The Word mail merge extension in Chapter 6 shows how the same data can also produce confirmation letters, renewal reminders, fee notices and ID cards without retyping. It can save a coordinator's time, give instant answers about members, dues and events, and be adapted by other clubs and societies.

7.2 Limitations

The workbook is sized for up to 100 members, 150 fee records, 20 events and 300 attendance rows, uses sample data and works for one user at a time. It does not send reminders or emails automatically.

7.3 Future Scope

- Add PivotTables and slicers for interactive analysis.
- Add an expense tracker and event budget analysis.
- Add a printable membership card or fee receipt for every member.
- Move the workbook to a shared online location so that several coordinators can update it.

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